

Multiplexer and Demultiplexer Tree

Multiplexer Tree Concept

- Large MUX built using smaller MUX
- Hierarchical structure (tree form)
- Reduces design complexity
- Example: 8:1 using 2:1 MUX

8:1 MUX using 2:1 MUX – Stage 1

- 4 multiplexers at first level
- Select between input pairs
- $I_0 I_1 \rightarrow M_0$
- $I_2 I_3 \rightarrow M_1$
- $I_4 I_5 \rightarrow M_2$
- $I_6 I_7 \rightarrow M_3$
- Select line: S_0

8:1 MUX using 2:1 MUX – Stage 2

- 2 multiplexers at second level
- $M0\ M1 \rightarrow M4$
- $M2\ M3 \rightarrow M5$
- Select line: $S1$

8:1 MUX using 2:1 MUX – Stage 3

- Final multiplexer
- M4 M5 → Output
- Select line: S2
- Implements full 8:1 selection

Demultiplexer Tree Concept

- Large DEMUX built using small DEMUX
- Hierarchical distribution
- One input $\rightarrow$ many outputs
- Example: 1:8 using 1:2 DEMUX

1:8 DEMUX using 1:2 DEMUX – Stage 1

- Single DEMUX at top
- Input $\rightarrow$ D0 D1
- Select line: S2

1:8 DEMUX using 1:2 DEMUX – Stage 2

- Two DEMUX at middle level
- $D0 \rightarrow Y0 Y1$
- $D1 \rightarrow Y2 Y3$
- Select line: S1

1:8 DEMUX using 1:2 DEMUX – Stage 3

- Four DEMUX at final level
- $Y_0 \rightarrow O_0 O_1$
- $Y_1 \rightarrow O_2 O_3$
- $Y_2 \rightarrow O_4 O_5$
- $Y_3 \rightarrow O_6 O_7$
- Select line: S_0

MUX Tree vs DEMUX Tree

- MUX: many inputs → one output
- DEMUX: one input → many outputs
- MUX selects data
- DEMUX distributes data
- Both use hierarchical design

Summary

- MUX tree builds large multiplexer
- DEMUX tree builds large demultiplexer
- Uses stages of smaller blocks
- Important digital design method